



**higher education
& training**

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA



KZN REGIONAL OFFICE

NATIONAL CERTIFICATE (VOCATIONAL)

FUNDAMENTALS MATHEMATICS NQF LEVEL 2

TASK: 6 (V1)

DATE: 12 AUGUST 2026

DURATION: 2h30min Controlled
Environment

MARKS: 50 **ICASS weighting:** 10

This question paper consists of 5 pages and a formula sheet.

SUBJECT OUTCOMES BEING ASSESSED:

SUBJECT OUTCOMES

3.1 - 3.5

LEARNING OUTCOMES

3.1.1 – 3.5.2

INSTRUCTIONS AND INFORMATION

1. Read **ALL** the questions carefully.
2. Answer **ALL** the questions.
3. Number the answers correctly according to the numbering system used in this question paper.
4. Write neatly and legibly.
5. The time allocated includes time for writing and completion of the assignment. The Keying in/completion of the assignment must be done in controlled environment
- 6 All work you do not want to be marked must be clearly crossed out.

STRON 1

Give the definitions of the following mathematical terms

2.1.1 Volume

(1)

2.1.2 Area

(1)

2.1.3 Angle

(1)

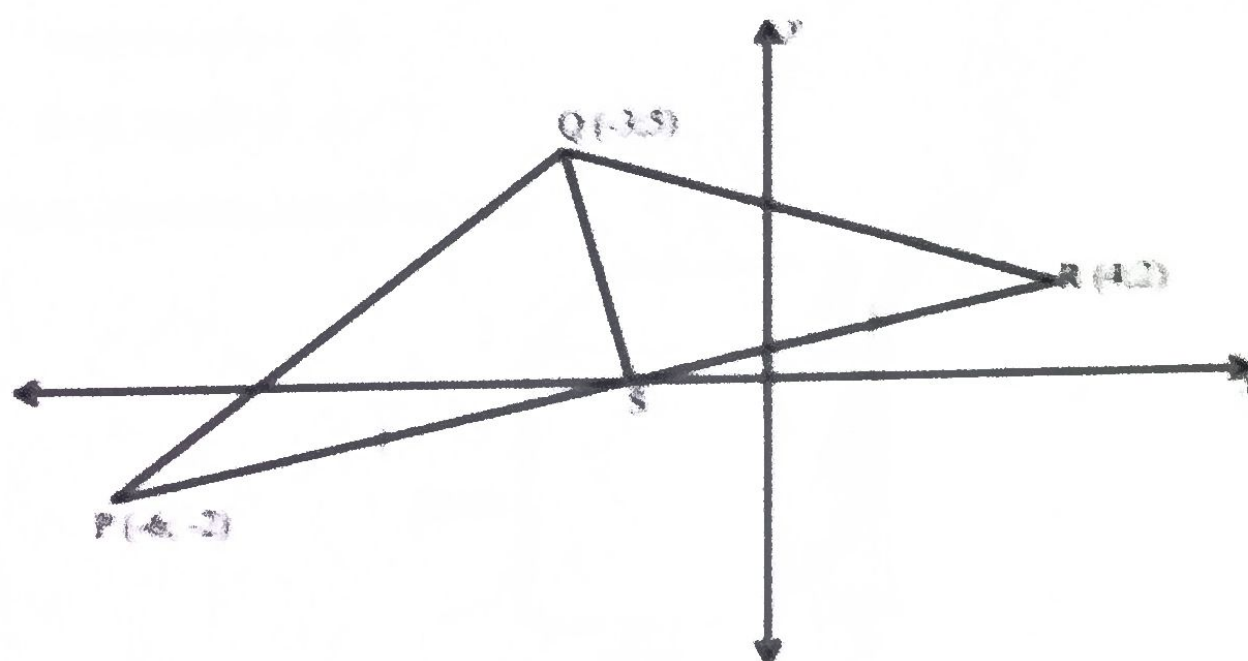
2.1.4 sphere

(1)

14

QUESTION 2

2.1 The vertices of $\triangle PQR$ are $P(-6,-2)$, $Q(-3,5)$ and $R(4,2)$. S is the midpoint of PR .



LEVEL: 2
DATE: 12
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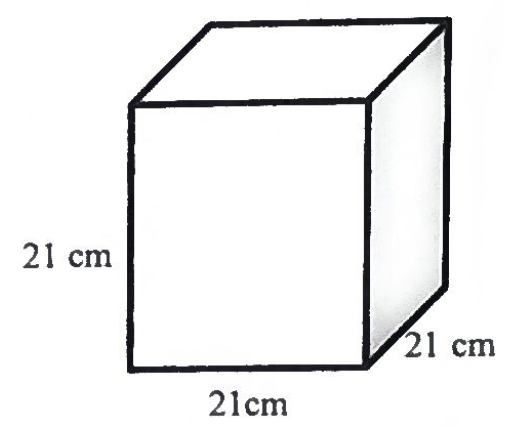
- 2.1.1 Calculate the coordinates of S. (2)
- 2.1.2 Calculate the length of line QR. (2)
- 2.1.3 Determine the gradient of line QR. (2)

2.2 V (a; -1) and W (5;3) are two points on a Cartesian plane.
If VW is 5 units in length, determine the value of a. (4)

- 2.3 Given $\triangle DEF$ with vertices D (2; -7); E (5; -2); F (4; -4).
 - 2.3.1 Find the image of $\triangle DEF$ by reflecting about $y = x$. (6)
 - 2.3.2 What would the coordinates of F be if it is translated 3 units up and 4 units to the left? (2)

- 2.4 Write down the rule used for each of the following reflections:
 - 2.4.2 A (3;4) to A' (-4; -3) (2)
 - 2.4.3 B (5; 2) to B' (5; -2) (2)

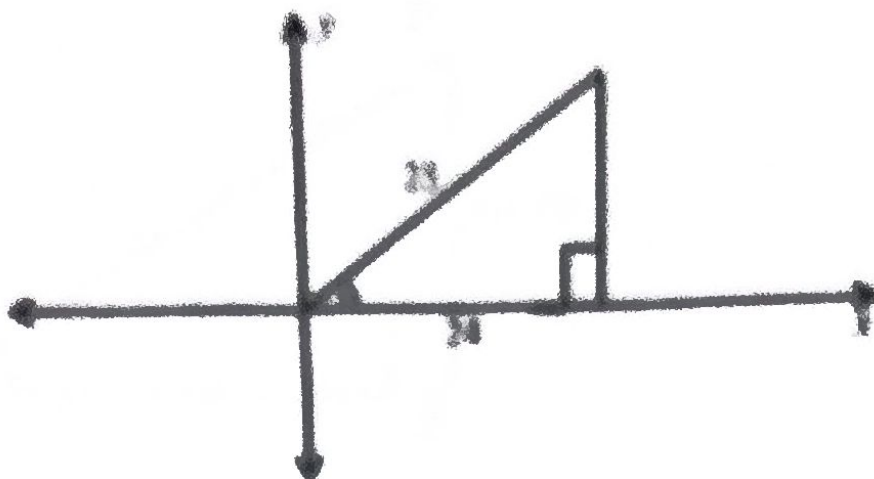
2.5 Given a closed cube with 21 cm sides:



- 2.5.1 Calculate the volume. (2)
- 2.5.2 Calculate the total surface area. (2)

[26]

If $\cos \theta = \frac{24}{25}$, $0^\circ \leq \theta \leq 90^\circ$ determine the following



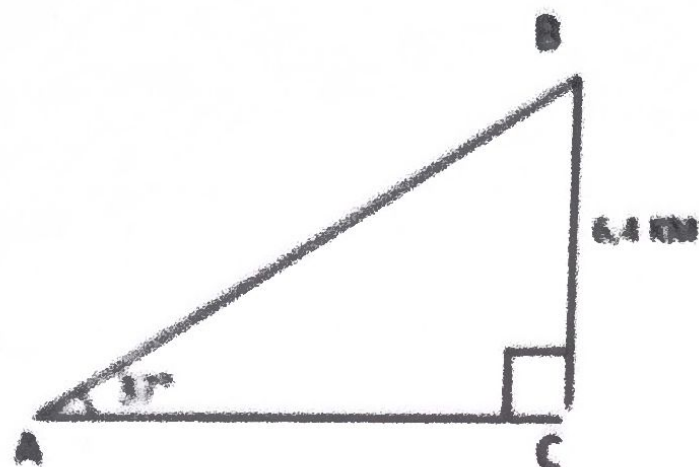
3.1.1 $\sin \theta$

(3)

3.1.2 $5 \cos \theta - 12 \tan \theta$

(3)

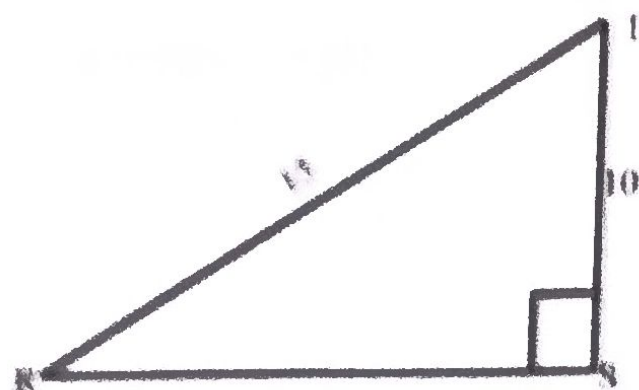
Q1. At a moment in time, BC is 6.4 km above the ground at point B. The angle of elevation from point A to point B is 37° . C is directly below the point B.



3.2.1 Calculate the distance AC. (3)

3.2.2 Calculate the distance AB. (3)

3.3 In triangle RST, RT = 15 units; ST = 10 units; $\hat{S} = 90^\circ$.



3.3.1 What is the name given to the longest side of the right-angled triangle (RT)? (1)

3.3.2 Write down $\cos \hat{T}$ as a ratio in its lowest form. (2)

3.3.3 Calculate $\hat{T}$. (2)

3.3.4 Calculate the length of RS. (3)

[20]

$$1) A_{\text{square}} = l \times l = l^2$$

$$2) A_{\text{rectangle}} = l \times w$$

$$3) A_{\text{triangle}} = \frac{1}{2} b \times h$$

$$4) A_{\text{circle}} = \pi r^2$$

$$5) C = 2\pi r$$

$$6) \text{Area of parallelogram} = \text{base} \times \text{perpendicular height}$$

$$7) A_{\text{hexagon}} = \frac{3\sqrt{3}}{2} L^2$$

$$8) A_{\text{hexagon}} = \frac{\sqrt{3}}{2} W^2$$

$$9) A_{\text{cylinder}} = 2\pi r(h+r)$$

$$10) \text{Volume} = \text{Area of base} \times \text{perpendicular height}$$

$$11) V_{\text{cube}} = l \times l \times l = l^3$$

$$12) V_{\text{cylinder}} = \pi r^2 h$$

$$13) \text{Total surface area cube} = 6 \times l^2$$

$$14) \text{Total surface area triangular prism} = (\text{height of prism} \times \text{perimeter of base}) + 2(\text{area of base})$$

$$15) m = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$16) D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$17) M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

$$18) \theta = \tan^{-1} m$$

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STUD $\bar{x} = \frac{\sum x}{n}$ or Mean = $\frac{\text{total or sum of all items}}{\text{number of items}}$

DA $R = X_n - X_1$ or Range = highest value - lowest value

21) $Q_{\text{position}} = \frac{j}{4}(n+1)$

22) Inter-quartile range = upper quartile - lower quartile = $(Q_3 - Q_1)$

23) Semi inter-quartile range = $\frac{1}{2}(\text{upper quartile} - \text{lower quartile}) = \frac{Q_3 - Q_1}{2}$

24) $P_{\text{position}} = \frac{j}{100}(n+1)$